

Oil Trading Commodity Multimodal Network Design Problem

Sets and Indices

- V_d : Set of demand nodes (origins/destinations)
- V_h : Set of potential hub nodes
- $V = V_d \cup V_h$ Set of all nodes
- T : Set of service types (e.g., rail, high-speed rail)
- $(i, j) \in V_d \times V_d, \quad i \neq j$ Commodity pairs (origin, destination)
- $k, m \in V_h$ Hub indices
- $t \in T$ Service type index

Parameters

- W_{ij} : Quantity of commodity from i to j
- C_{ij}^{road} : Cost of direct road transport from i to j
- $C_{i,k}^{access}$: Cost of transporting from origin i to hub k
- $C_{k,m,t}^{hub}$: Cost of transporting between hubs k and m with service t
- $C_{k,j}^{delivery}$: Cost of transporting from hub k to destination j
- F_k : Fixed cost to open hub k
- F_{kmt} : Fixed cost to activate service t between hubs k and m

Decision Variables

- $z_k \in \{0, 1\}$ 1 if hub k is selected
- $y_{kmt} \in \{0, 1\}$ 1 if service t is activated between hubs k and m
- $e_{ij} \in \{0, 1\}$ 1 if commodity (i, j) goes directly by road
- $a_{ijk} \in \{0, 1\}$ 1 if commodity (i, j) accesses hub k from origin i
- $x_{ijkmt} \in \{0, 1\}$ 1 if (i, j) goes from hub k to m using service t
- $b_{ijk} \in \{0, 1\}$ 1 if commodity (i, j) is delivered from hub k to j

Objective Function

$$\begin{aligned} \min \quad & \sum_{i,j} W_{ij} \cdot \left[C_{ij}^{road} \cdot e_{ij} + \sum_k C_{i,k}^{access} \cdot a_{ijk} + \sum_{k,m,t} C_{k,m,t}^{hub} \cdot x_{ijkmt} + \sum_k C_{k,j}^{delivery} \cdot b_{ijk} \right] \\ & + \sum_k F_k \cdot z_k + \sum_{k,m,t} F_{kmt} \cdot y_{kmt} \end{aligned}$$

Constraints

1. Service Activation Implies Hub Open

$$\begin{aligned} y_{kmt} &\leq z_k \quad \forall k, m \in V_h, t \in T \\ y_{kmt} &\leq z_m \quad \forall k, m \in V_h, t \in T \end{aligned}$$

2. Commodity Must Choose a Unique Route

$$e_{ij} + \sum_k a_{ijk} = 1 \quad \forall i, j \in V_d, i \neq j$$

3. Access Flow Implies Hub is Open

$$a_{ijk} \leq z_k \quad \forall i, j \in V_d, k \in V_h$$

4. Delivery Flow Implies Hub is Open

$$b_{ijk} \leq z_k \quad \forall i, j \in V_d, k \in V_h$$

5. Flow Uses Only Active Links

$$x_{ijkmt} \leq y_{kmt} \quad \forall i, j \in V_d, k, m \in V_h, t \in T$$

6. Flow Conservation for Multimodal Paths

$$\begin{aligned} \sum_k a_{ijk} &= \sum_{k,m,t} x_{ijkmt} \quad \forall i, j \in V_d, i \neq j \\ \sum_{k,m,t} x_{ijkmt} &= \sum_k b_{ijk} \quad \forall i, j \in V_d, i \neq j \end{aligned}$$